Here's the Kicker

Correcting Selection Bias in NFL Field Goal Models via Inverse Probability Weighting

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Coaches choose which kicks we see

Field goal models adjust attempts for difficulty. Distance first, then weather, venue, game state.

All of them are fit on kicks a coach agreed to take.

Coaches decline potential kicking opportunities in favor of punting or going for it on fourth down.

These decisions are not random. Actual attempts represent a selected group, overrepresenting favorable kicking scenarios.

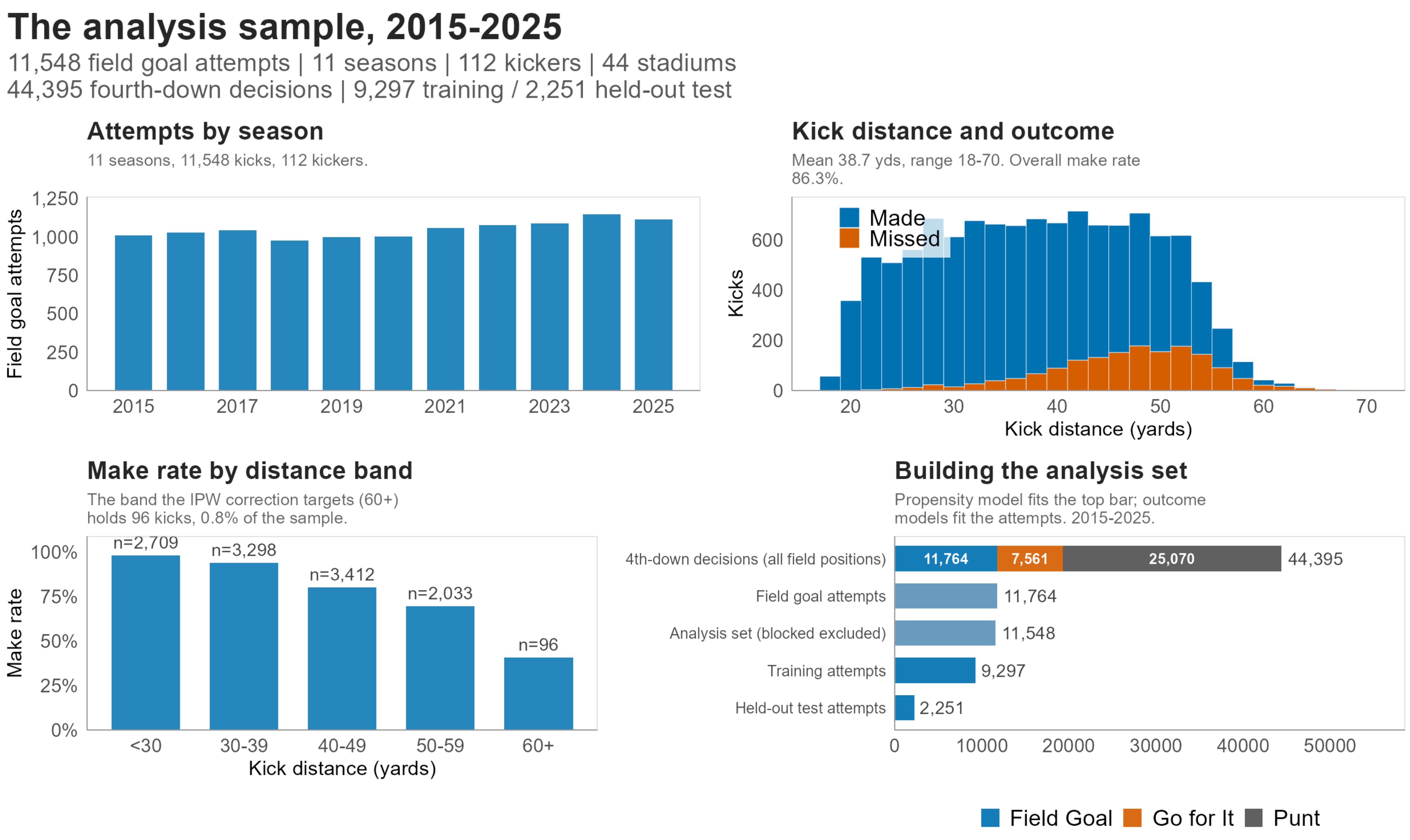
> If a coach believes they have information about their kicker's ability, that belief can shape which kicks get attempted, information no outcome model observes directly.

Three questions

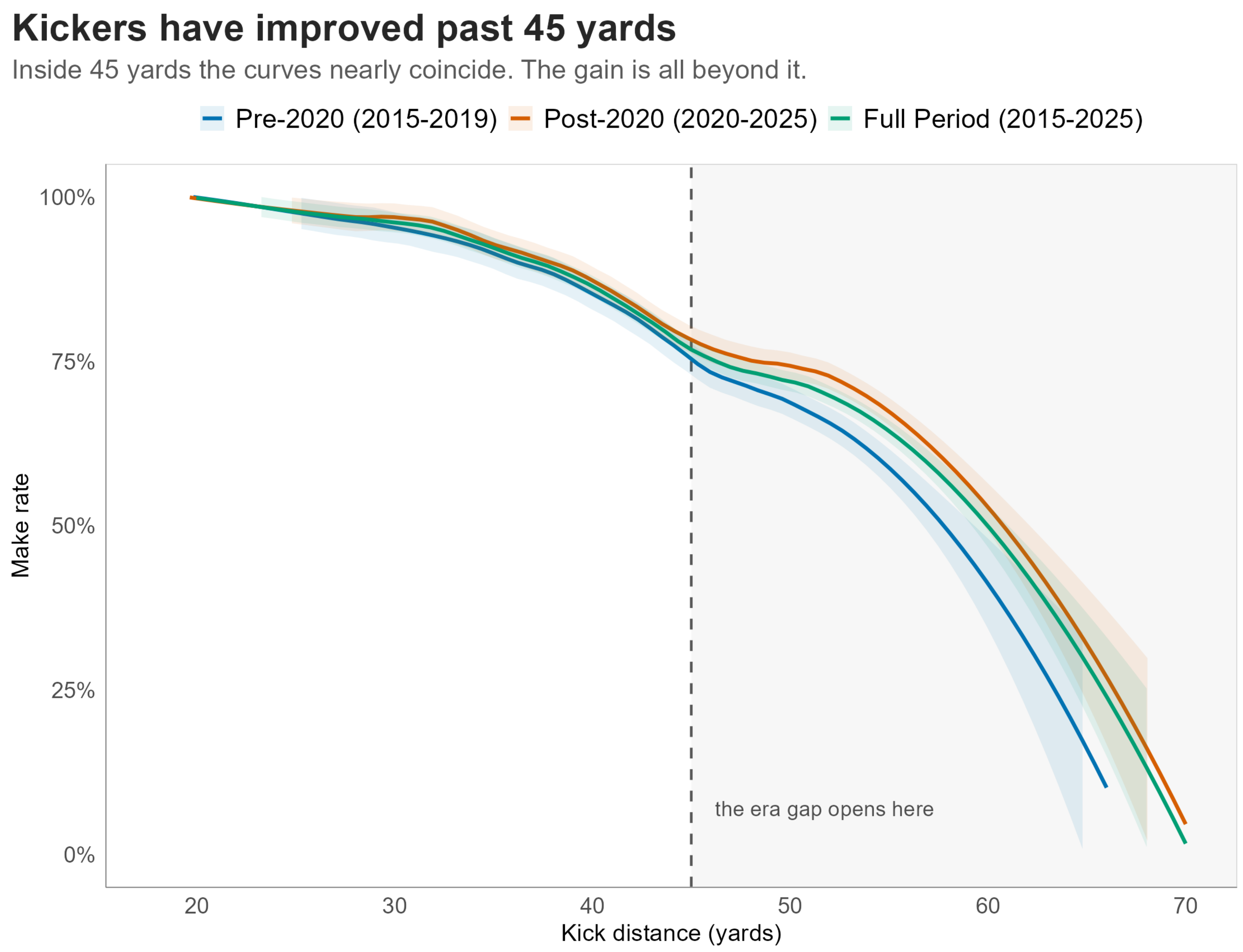
1. Is the attempt decision **structured enough** to model?
2. Does adding the declined kicks back with **pseudo-miss labels** help?
3. Does **inverse probability weighting** improve out-of-sample prediction?

> And if the answer to two and three is no, what does that tell us about why coaches decline?

The analysis sample



Kickers have improved past 45 yards



~202 additional makes

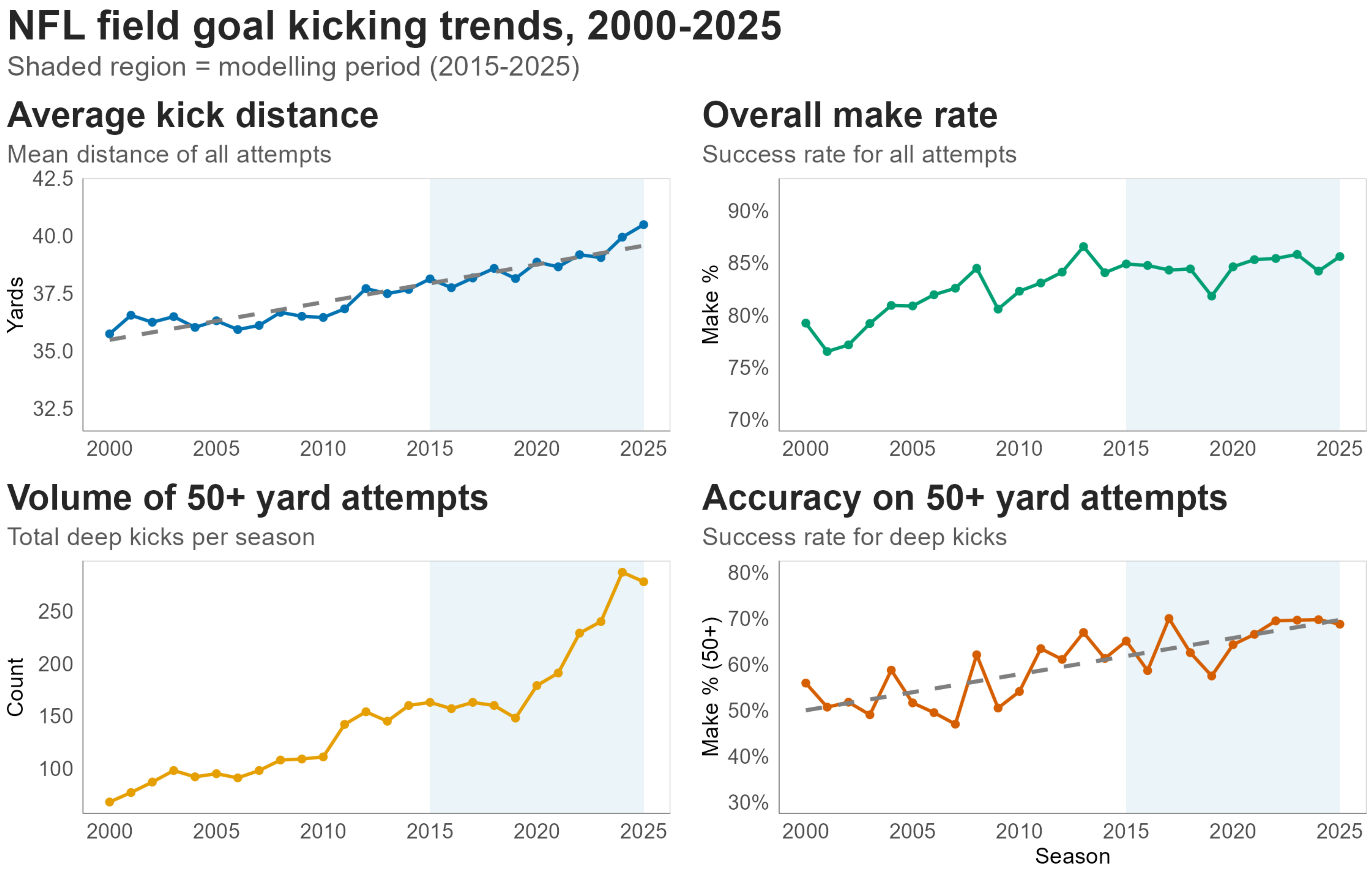
in 2020 to 2025, against the pre-2020 success curve

~34 makes per season

and ~148 of the ~202 come from beyond 45 yards

> Inside 45 yards the curve was already close to its ceiling. The improvement is concentrated in the long tail.

Coaches are asking for longer kicks



Four outcome models

M0 distance only

B-spline in distance. Nothing else.

The predictive floor.

M1 full GLMM

M0 plus weather splines, contextual indicators, kicker-season and stadium random intercepts. Unweighted.

The baseline.

M2 augmented

M1 fitted on an augmented set. PATs with real outcomes, declined fourth downs as pseudo-misses, both at reduced weight.

Impute the unobserved attempts.

M3 IPW

M1 fitted with stabilized IPW case weights.

Reweight instead of impute.

Every row carries an observed outcome. The price is variance, not bias.

Modeling the attempt decision

Multinomial logit over kick, punt, go. Reference category is go for it.

- Natural spline on **field position** yl_i, full field 1 to 100
- **Log game seconds remaining** $\log(1 + t_i)$, quarter-time interactions
- Yards to go ytg_i; score state, venue, wind, temperature $\mathbf{S}_i$
- **Leverage** L_i, the win probability after a make minus the win probability after a miss, estimated before the outcome is known

Rolling origin. Train on seasons $s-3$, $s-2$, $s-1$, predict season s.

> **AUC 0.933 to 0.959** in plausible kicking range, every season
>
> **RMSE 0.045** against realized win probability, after endgame overrides

$$\log \frac{P(D_i = d)}{P(D_i = \mathrm{Go})} = \alpha_d + \sum_{k=1}^{6} \beta_{d,k}\, n_k^{\mathrm{fp}}(\mathrm{yl}_i) + \gamma_d \log(1 + t_i) + \delta_d\, \mathrm{ytg}_i + \boldsymbol{\theta}_d^{\top} \mathbf{S}_i + \psi_d L_i$$

From propensities to weights

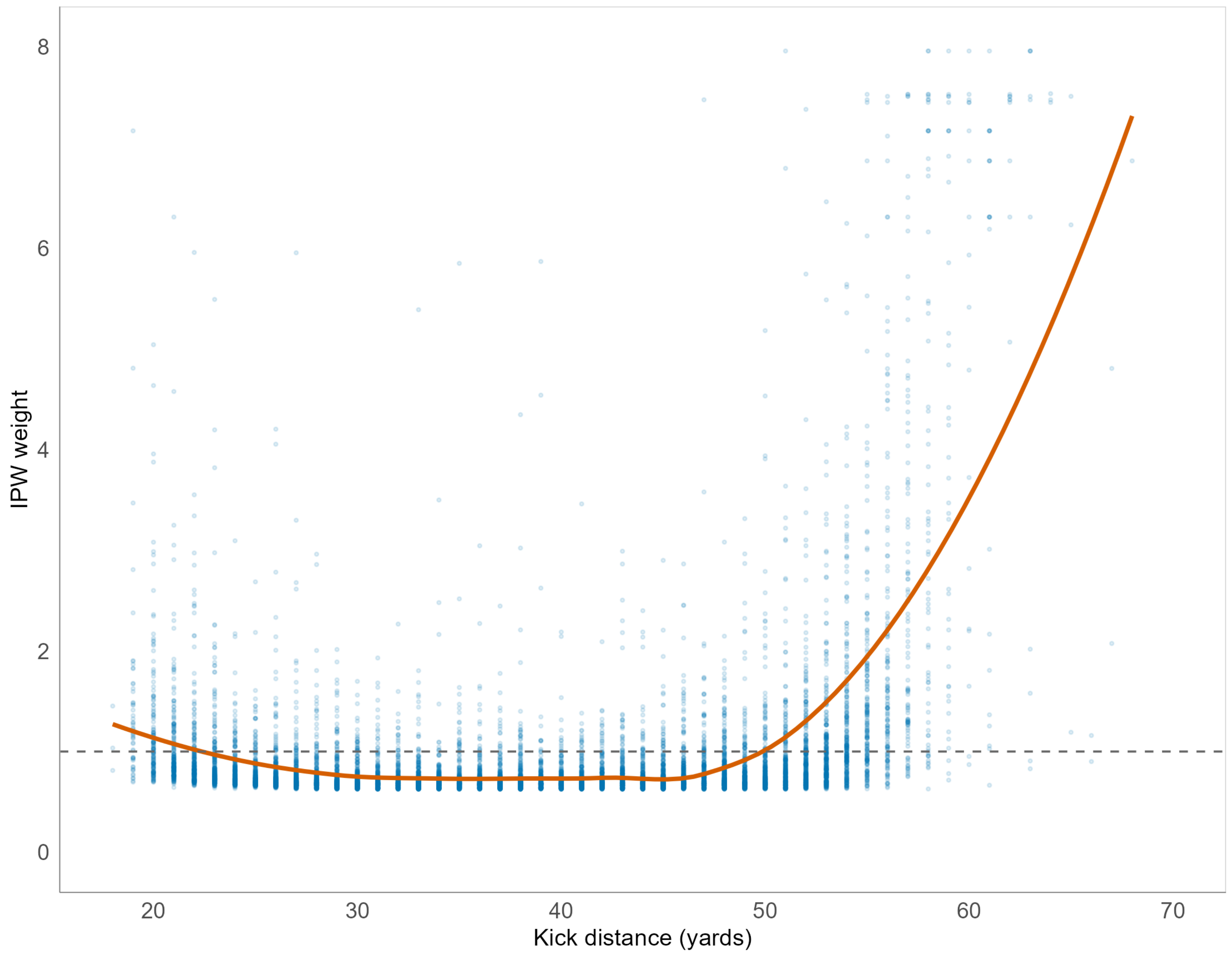
$$w_i = \frac{\bar{\pi}}{\hat{\pi}(X_i)}$$

- Propensities $\hat{\pi}(X_i)$ clipped to 0.02 and 0.98 before inversion
- w_i Hajek-normalized to mean 1, capped at three seasonal standard deviations, renormalized
- w_i normalized over **kickable** situations only

> Rare, hard attempts are upweighted so the observed sample better represents the full kickable population. Final weights span 0.63 to 8.49.

The weights load onto the long tail

Stabilised, Hajek-normalised IPW weight by distance. Dashed line = 1.



The outcome model

- Seven-column B-spline in distance (d_i). Interior knots 28, 38, 48, 58. Boundaries 18 and 70.
- Natural splines, 3 df each, in standardized wind (W_i), temperature (T_i), and humidity (H_i).
- Contextual indicators ($\mathbf{C}_i$). Turf, precipitation, clock running, last two minutes, iced, playoffs.
- Kicker-season random intercept $u_{s(i)}$. Stadium random intercept $v_{g(i)}$.

$$\operatorname{logit} P(\mathrm{make}_i) = \sum_{k=1}^{7} \beta_k b_k(d_i) + \sum_j \gamma_j n_j(W_i, T_i, H_i) + \boldsymbol{\delta}^\top \mathbf{C}_i + u_{s(i)} + v_{g(i)}$$

> Identical specification for M1, M2 and M3; only the training set and the case weights change.

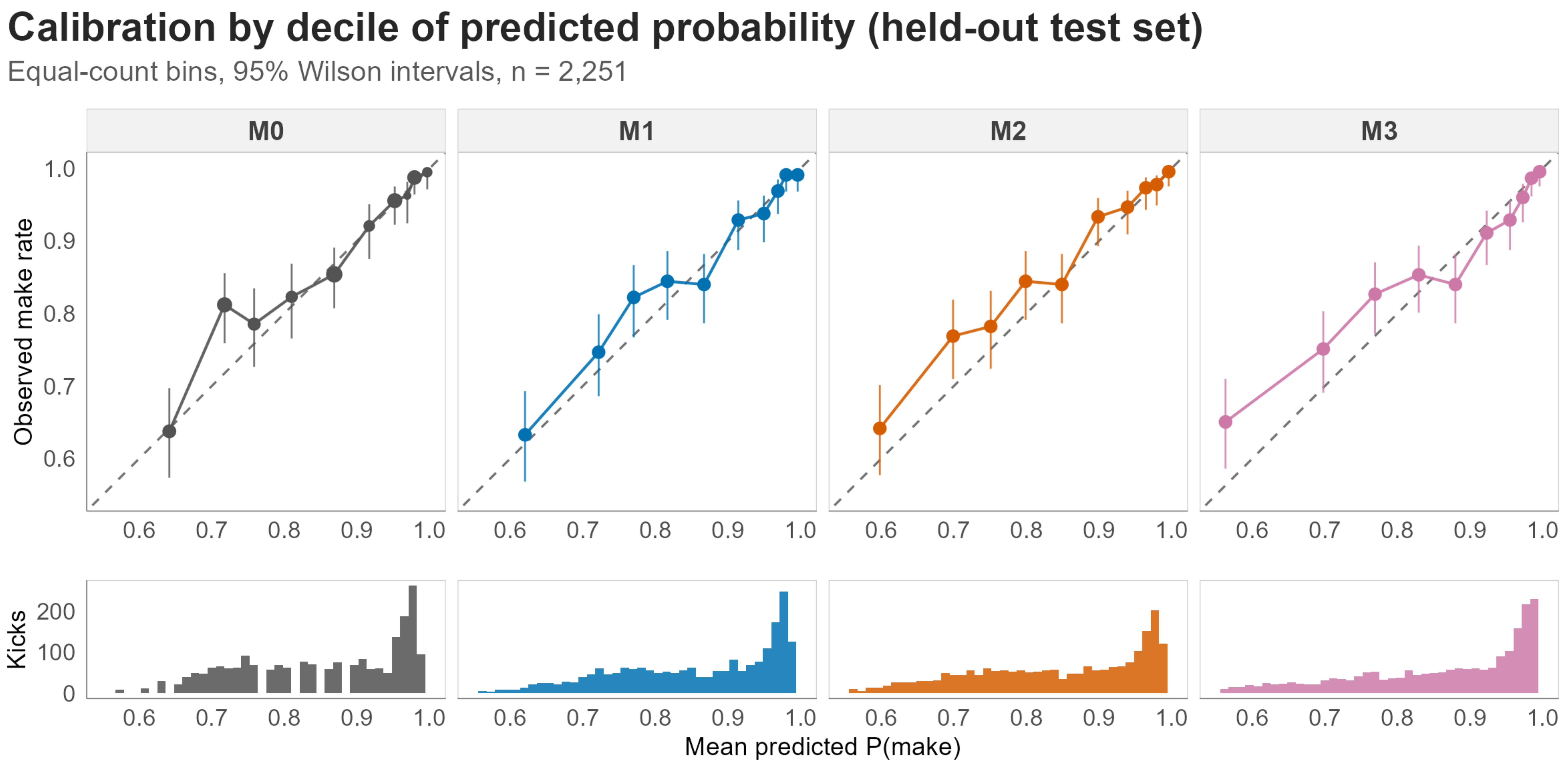
The ordering reverses

Model	IS Brier	OOS Brier	IS Log-Loss	OOS Log-Loss
M0 distance only	0.1045	0.1012	0.3387	0.3323
M1 full GLMM	0.1004	**0.1004**	0.3259	**0.3295**
M2 augmented	0.1009	0.1013	0.3281	0.3323
M3 IPW	**0.0982**	0.1030	**0.3216**	0.3383

In-sample $n = 11{,}548$. Held-out $n = 2{,}251$ from 593 disjoint games. Lower is better. AUC orders identically.

> **M3 wins in-sample on every metric and loses out-of-sample on every metric.** Out of sample it is outperformed by the distance-only model.

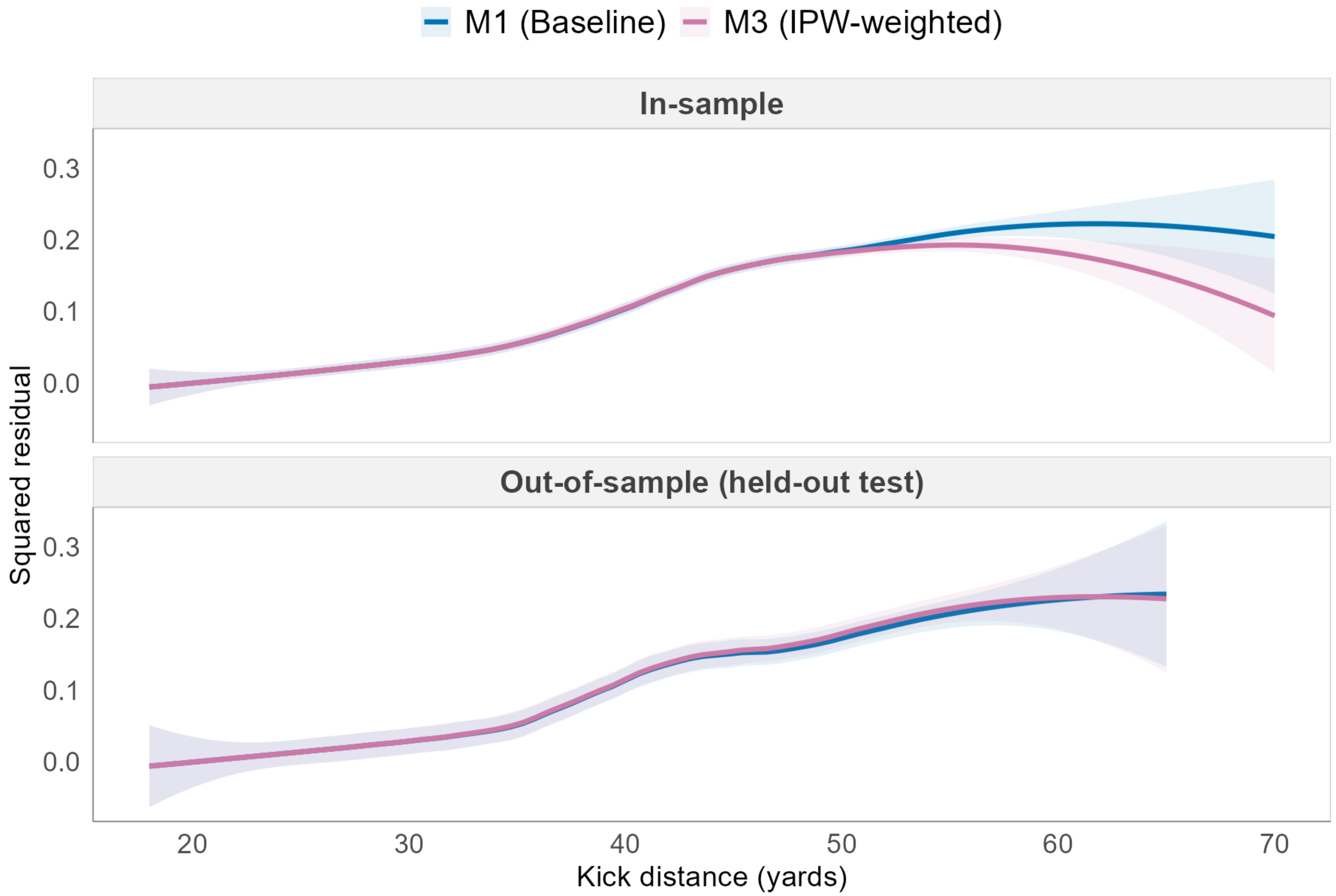
Calibration on held-out kicks



Why neither correction wins

The correction works where it was designed to, and only there

LOESS squared error by distance. In-sample M3 wins the long tail; out-of-sample the curves cross.



M1 Brier: IS 0.1004 / OOS 0.1004. M3 Brier: IS 0.0982 / OOS 0.1030.

M2. The labels are systematically wrong.

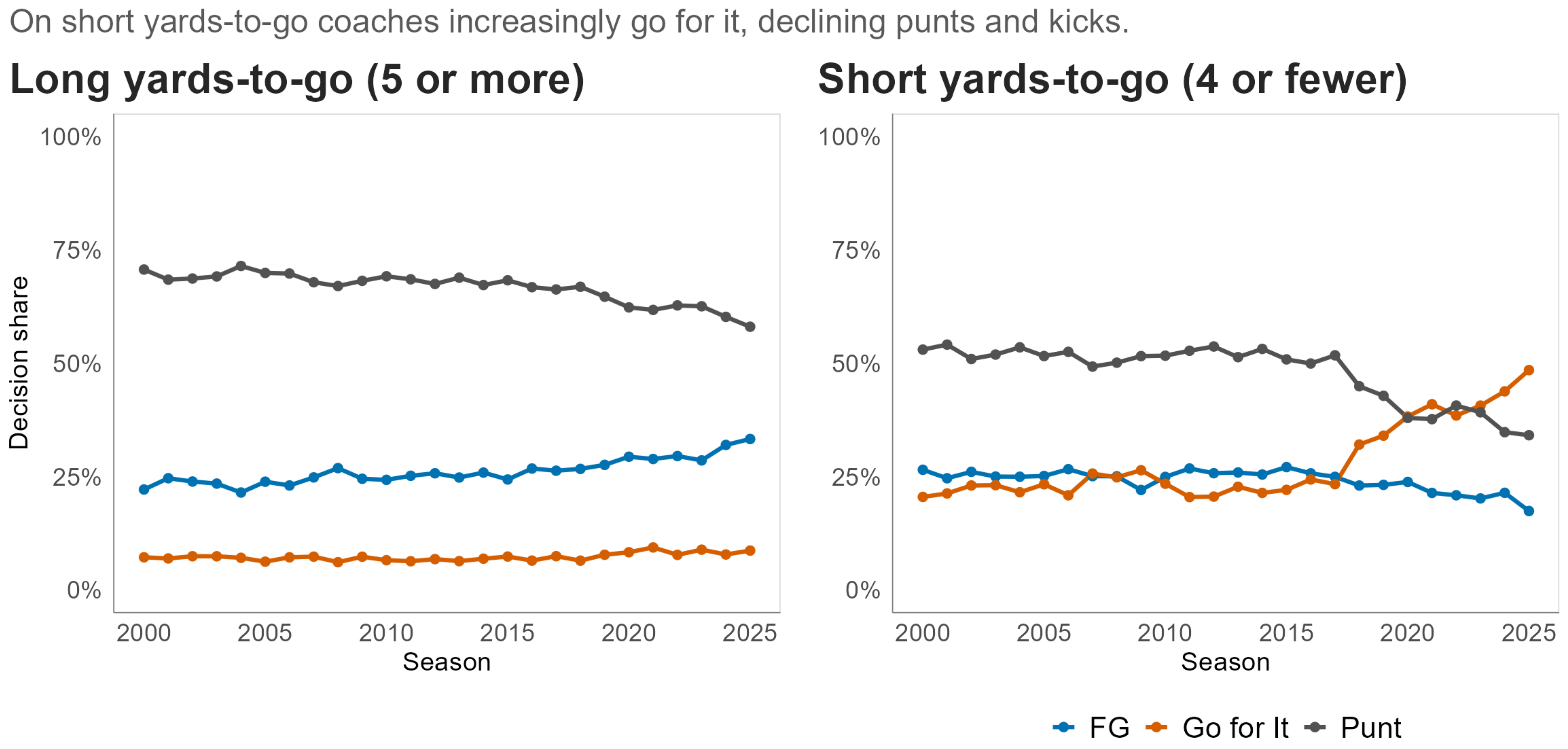
Every declined kick beyond 33 yards is entered as a miss. Degradation is monotone in the augmentation weight.

Aug. weight	IS Brier
0.05	0.1007
0.10	0.1009
0.25	0.1013

M3. The weights concentrate fit on the sparsest part of the curve.

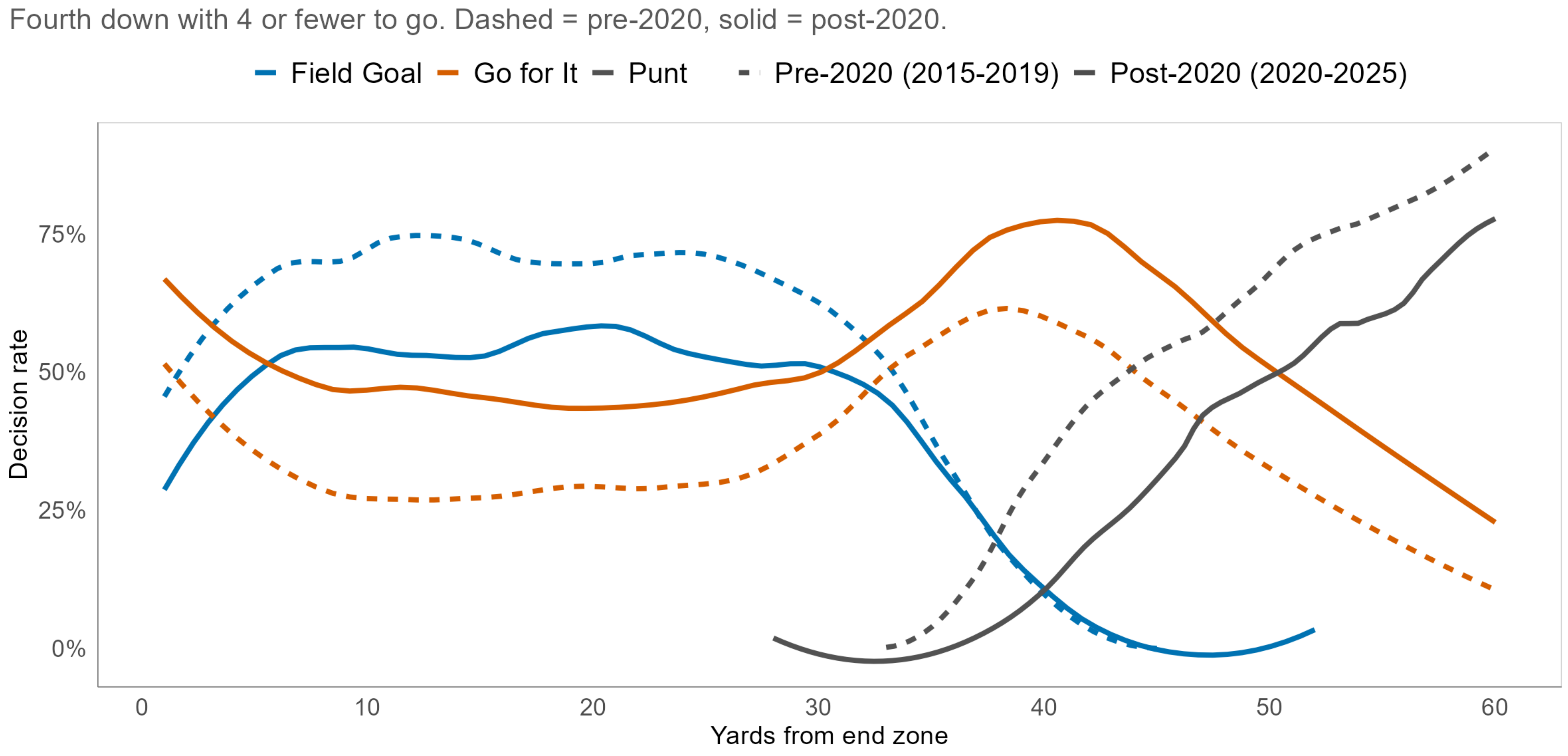
Weight is concentrated on long, low-propensity attempts. In-sample error drops exactly there. Out of sample the curves cross.

To kick or not to kick



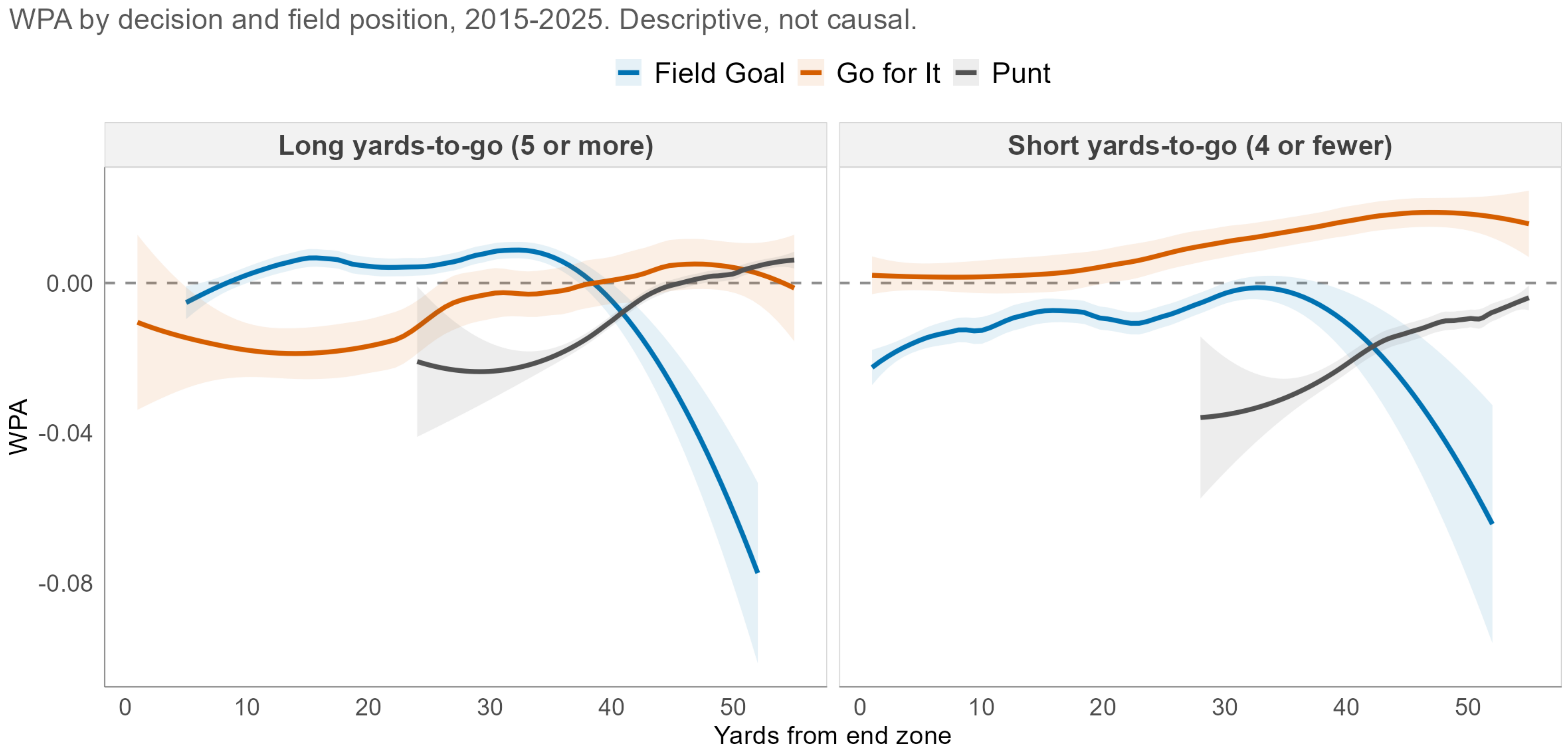
On long yards to go, field goals keep taking share from punts, in step with kicker improvement. On short yards to go, going for it surges after about 2018, a shift better kicking does not explain.

Fourth and short, by era



> Post-2020, coaches go for it more at nearly every field position. If declines were about doubting the kicker, improving kickers should have moved these curves the other way.

The value was in the alternative



On short yards to go, going for it carries positive win probability added almost everywhere, while a field goal does not. The decline looks less like doubt about the kicker and more like taking the higher-value option.

What we take from this

Findings

1. Coaching selection is strongly structured. In-range AUC 0.93 to 0.96 every season.
2. Pseudo-label augmentation fails, and fails worse the harder you lean on it.
3. IPW corrects what it claims to in-sample at long distance and still does not generalize. M1 is the better forecaster in both eras.

Recommendation

For in-game make probability, use the unweighted GLMM.

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What that appears to say about the mechanism

We assumed coaches decline out of doubt about the kicker. The decision curves and the WPA landscape point elsewhere. If the decision carries little information about kicker ability, there is little for a correction to recover.

Next

Kicker evaluation via an over-expected model. Coaching analysis via a win-probability-maximizing decision model. A causal treatment of the attempt decision.

Thank you

Questions welcome.

Data via **nflfastR**.

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Key references

Pasteur and Cunningham-Rhoads, on distance-adjusted kicker evaluation.

Osborne and Levine, on shrinkage estimation of field goal success.

Horvitz and Thompson, on sampling without replacement from a finite universe.

Hernan and Robins, *Causal Inference*, on stabilized weights.

New Diamond Open Access journal

Journal of Statistics and Data Science in Sports

https://journals.charlotte.edu/index.php/jsdss

Train and test balance

APPENDIX A1

Distance Band	Train n	Train share	Test n	Test share	Diff (pp)
< 30	2,214	23.8%	495	22.0%	-1.82
30-39	2,654	28.5%	644	28.6%	+0.06
40-49	2,716	29.2%	696	30.9%	+1.71
50-59	1,636	17.6%	397	17.6%	+0.04
60+	77	0.8%	19	0.8%	+0.02
Mean kick distance		38.68		39.00	+0.32 yds

Season-stratified game-level permutation test on the band shares, 2,000 redraws. Observed total variation distance 0.0182, null median 0.0162, $p = 0.385$. The realized split is an unremarkable draw.

Era stability of the result

APPENDIX A2

Era	M1 Brier	M3 Brier	M1 Log-Loss	M3 Log-Loss
Pre-2020 (n = 976)	**0.1105**	0.1141	**0.3583**	0.3702
Post-2020 (n = 1,275)	**0.0928**	0.0945	**0.3074**	0.3139

> M1 beats M3 in both halves, on both scoring rules. The result is not a post-2020 artifact.

Full results

APPENDIX A3

Model	IS Brier	IS Log-Loss	IS AUC	OOS Brier	OOS Log-Loss	OOS AUC
M0 distance only	0.1045	0.3387	0.774	0.1012	0.3323	0.762
M1 full GLMM	0.1004	0.3259	0.804	**0.1004**	**0.3295**	**0.769**
M2 augmented	0.1009	0.3281	0.800	0.1013	0.3323	0.766
M3 IPW	**0.0982**	**0.3216**	**0.805**	0.1030	0.3383	0.757

In-sample $n = 11{,}548$. Held-out $n = 2{,}251$. Brier and log-loss lower is better, AUC higher is better. Blocked kicks excluded.

Where exactly M3 improved

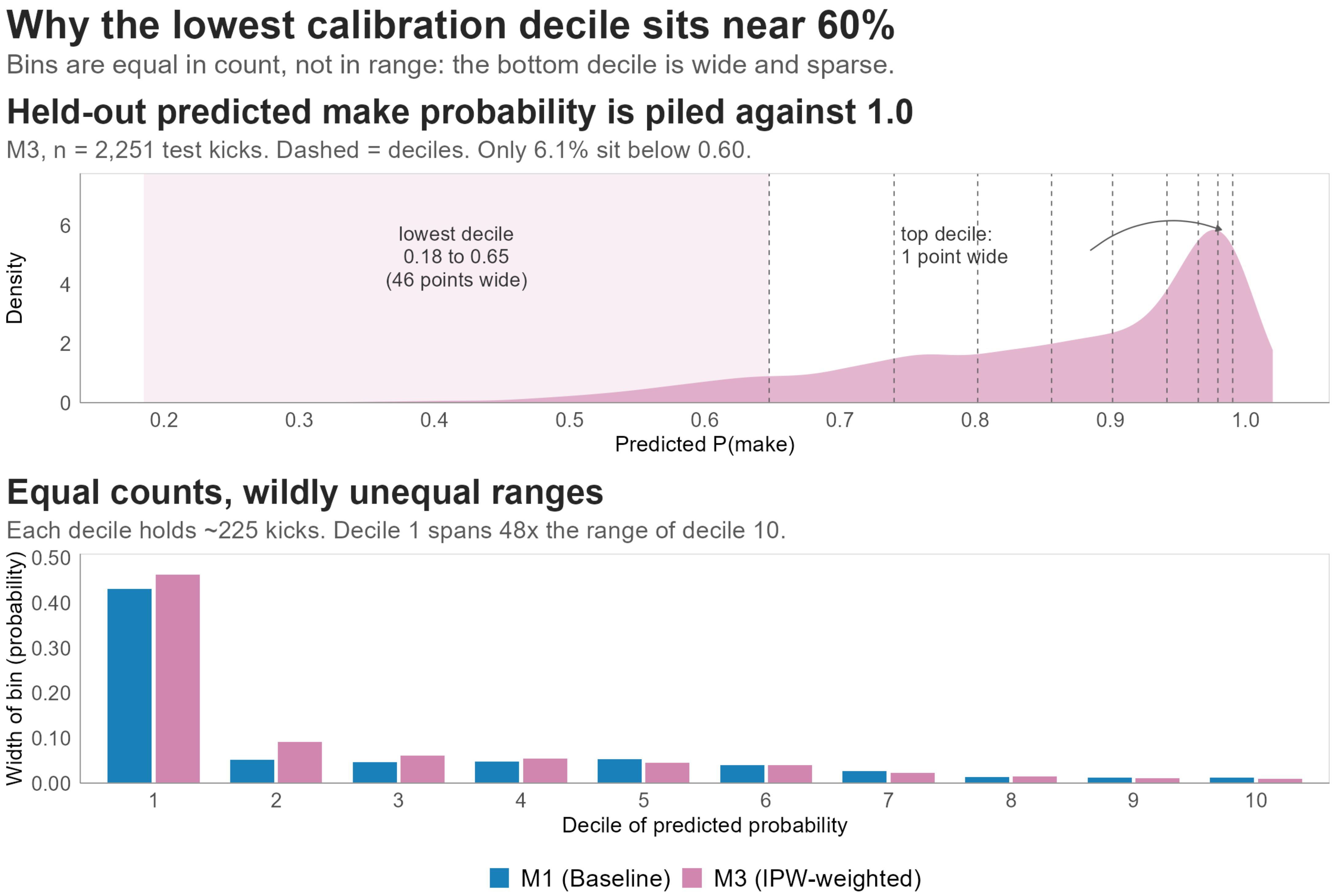
APPENDIX A4

Distance Band	n	M0	M1	M2	M3
< 30	2,709	0.0162	0.0161	0.0161	0.0161
30-39	3,298	0.0553	**0.0543**	0.0545	0.0545
40-49	3,412	0.1563	**0.1502**	0.1508	0.1503
50-59	2,033	0.2090	0.1981	0.1996	**0.1879**
60+	96	0.2298	0.2188	0.2193	**0.1657**

In-sample Brier by distance band. Bands are half-open, so 60+ means 60 yards or more. M3 gains 5.2% at 50-59 and 24.3% at 60+ versus M1.

Why the lowest decile sits near 60%

APPENDIX A5



Weight construction in full

APPENDIX A6

Three stages, per season

1. Stabilize. $w_i = \bar{\pi}/\hat{\pi}(X_i)$, with $\hat{\pi}$ clipped to $[0.02, 0.98]$ before inversion.
2. Hajek-normalize to mean 1, then cap at $\bar{w}_s + 3\sigma_{w,s}$, then renormalize.
3. $\bar{\pi}$ and both normalizations are computed over the **in-range** subset, not the full propensity frame.

What the cap costs and buys

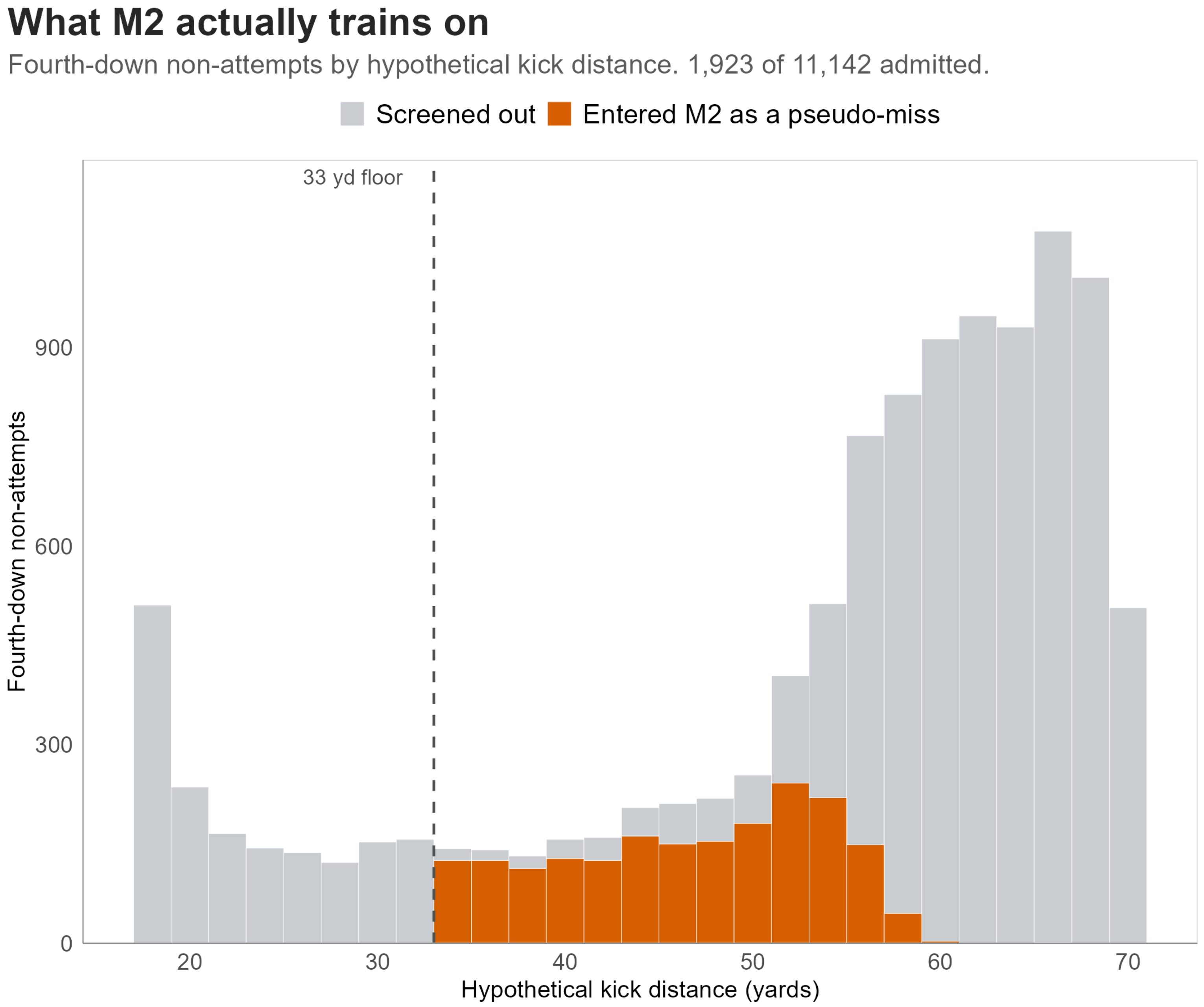
Before capping, one attempt carried a case weight of **1,633**.

After capping, weights span **0.63 to 8.49**, median 0.71.

Effective sample size falls from 11,764 to about **6,300**, so the reweighting costs roughly 46% of the sample.

M2 inclusion criteria

APPENDIX A7



Primary screen: propensity at least 0.25 and distance beyond 33 yards. 1,923 of 11,142 candidates admitted, with pseudo-miss labels biased in a known direction.

Propensity floor	Pool	OOS Brier
M1, no augmentation	0	0.1004
0.50	1,087	0.1009
0.35	1,543	0.1012
0.25 (primary)	1,923	0.1013
0.15	2,470	0.1016
0.10	2,903	0.1018

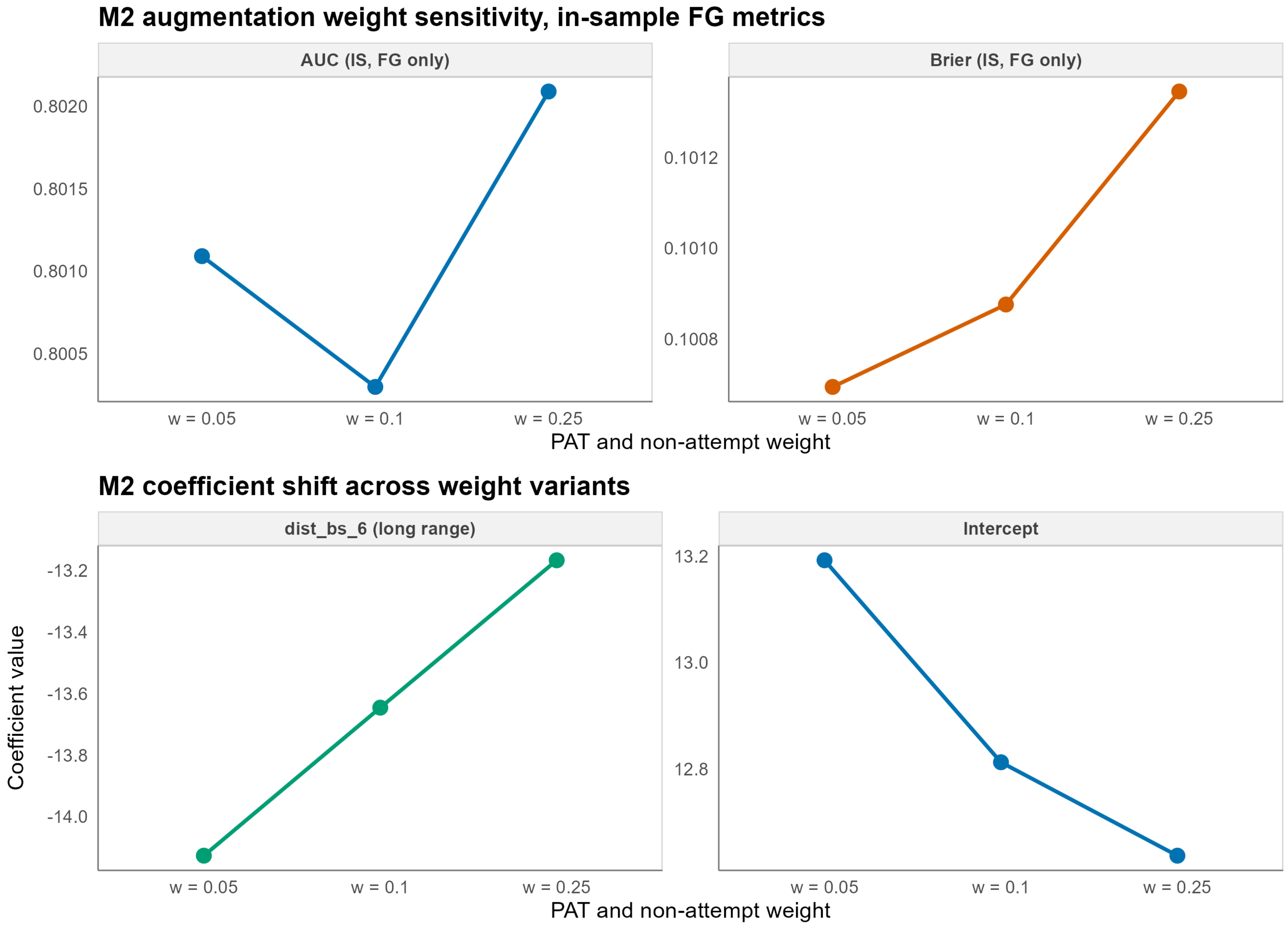
Monotone, and in the unhelpful direction. Every step down in the floor makes it worse, and the limit of the sweep is no augmentation at all.

M2 weight sensitivity in full

APPENDIX A8

M2 augmentation sensitivity, PAT and non-attempt weight in {0.05, 0.10, 0.25}

Metrics are FG-only in-sample, and the model is robust across this weight range.

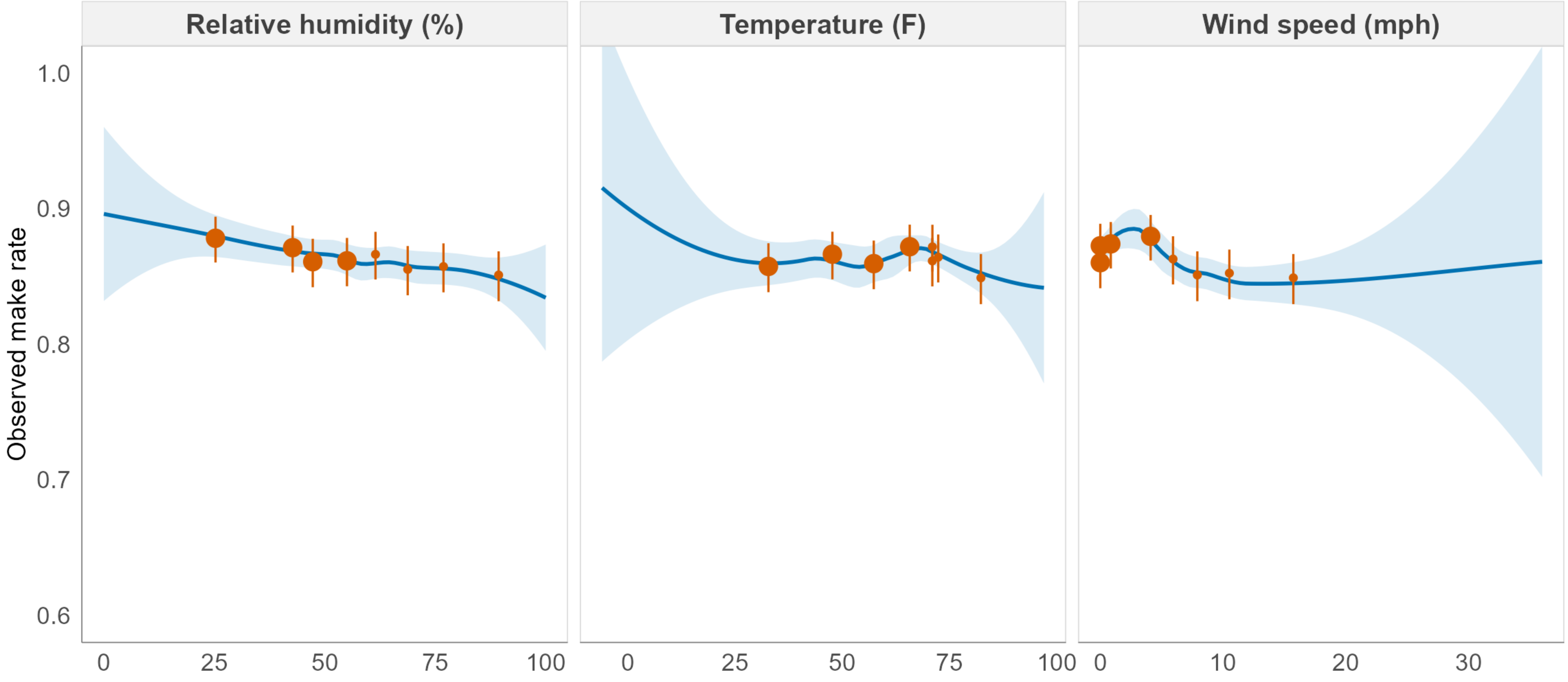


Weather covariates, raw

APPENDIX A9A

Raw make rate against the weather covariates

Orange: equal-count bins, 95% Wilson. Blue: LOESS. Not adjusted for distance.

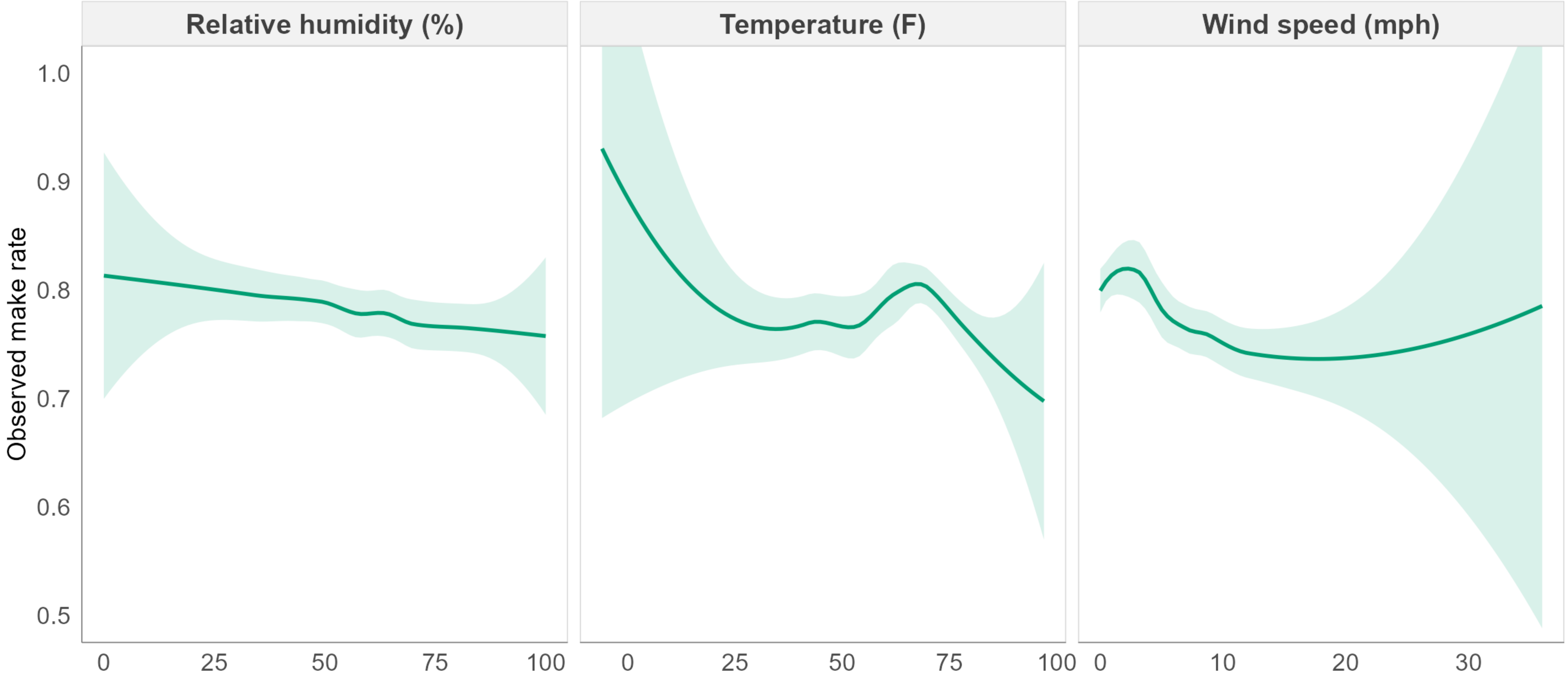


Weather covariates, distance held

APPENDIX A9B

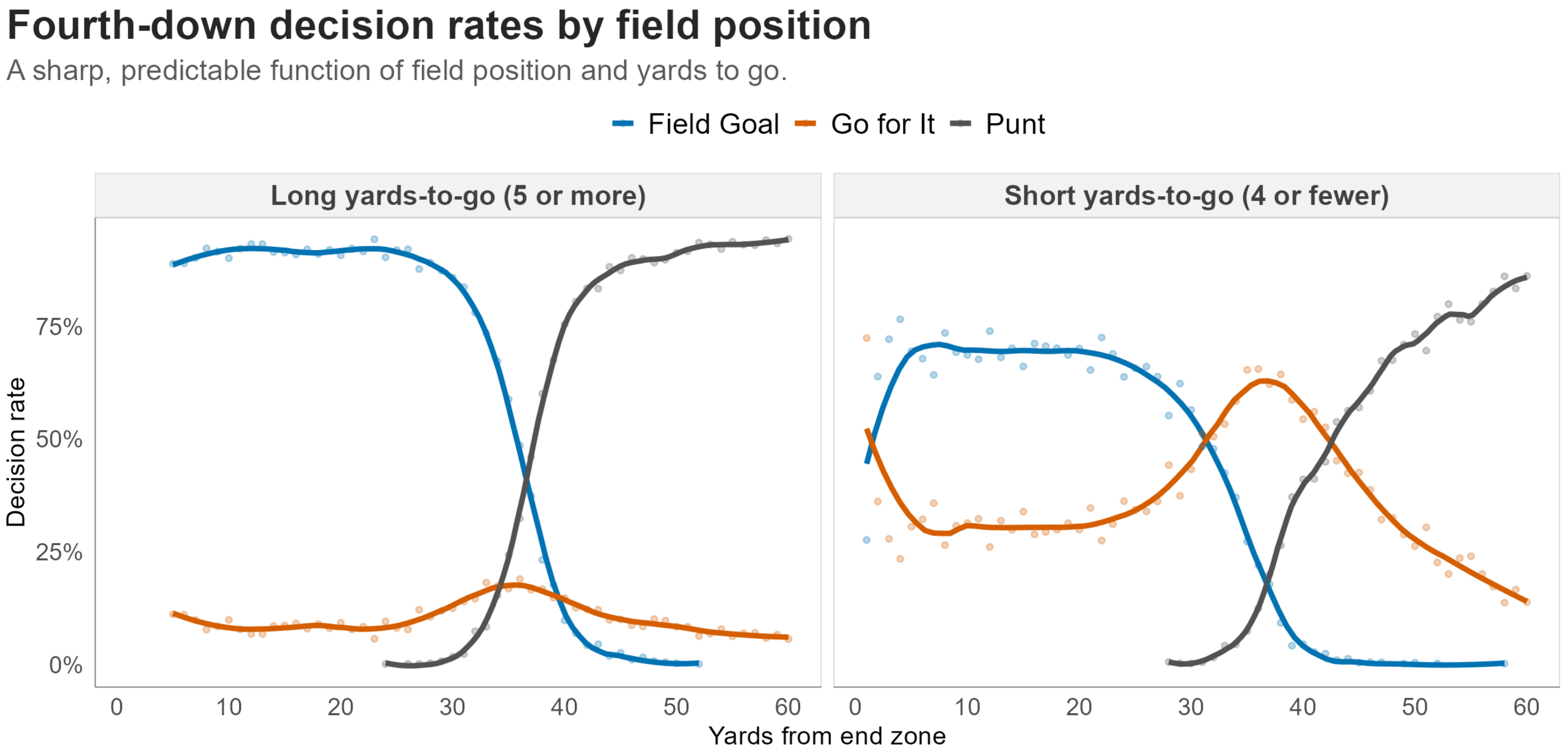
The same covariates, holding distance roughly fixed

40-54 yard attempts only. Removes the distance mix that confounds the panel above.



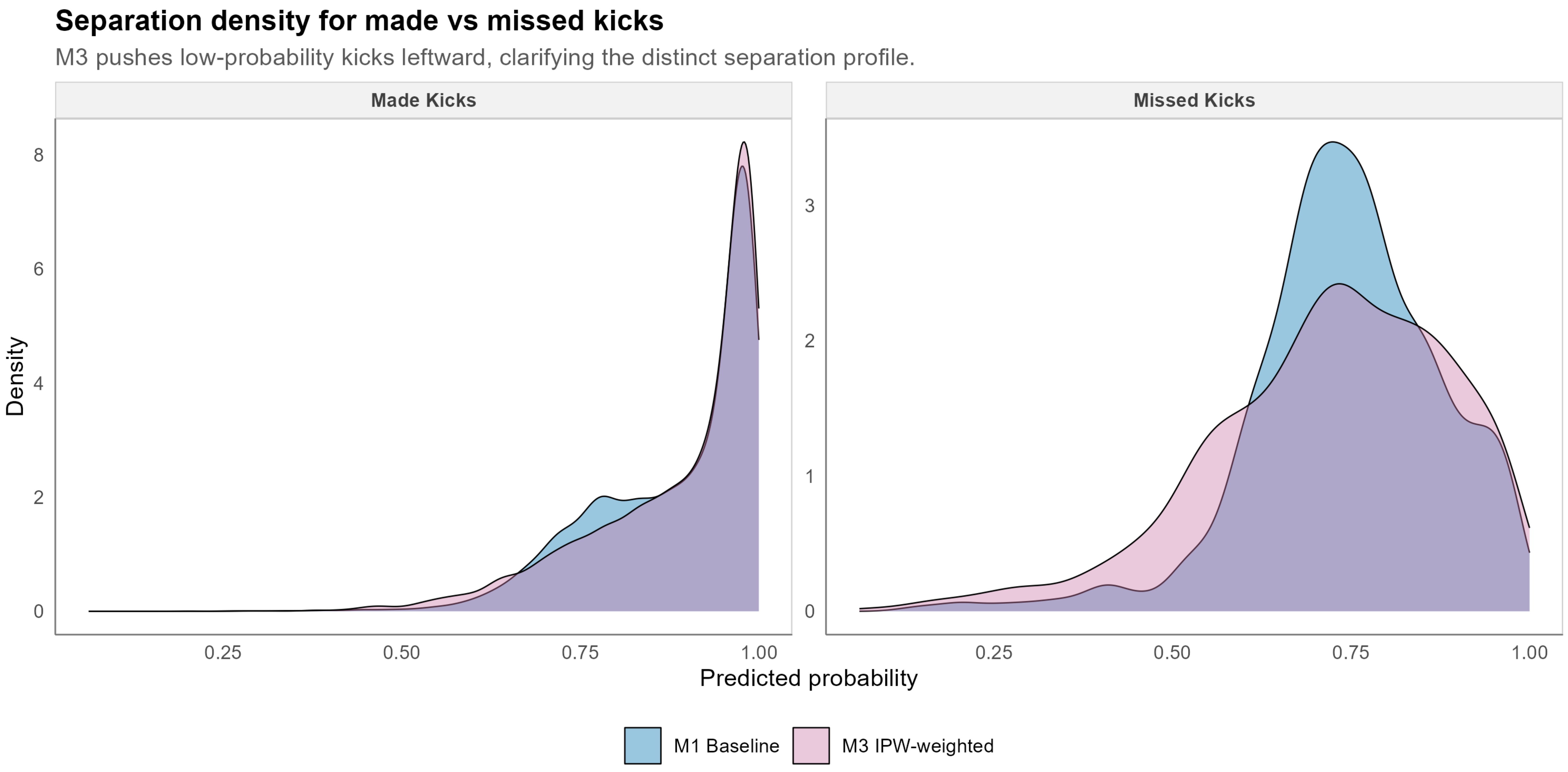
Fourth-down decisions by field position

APPENDIX A10



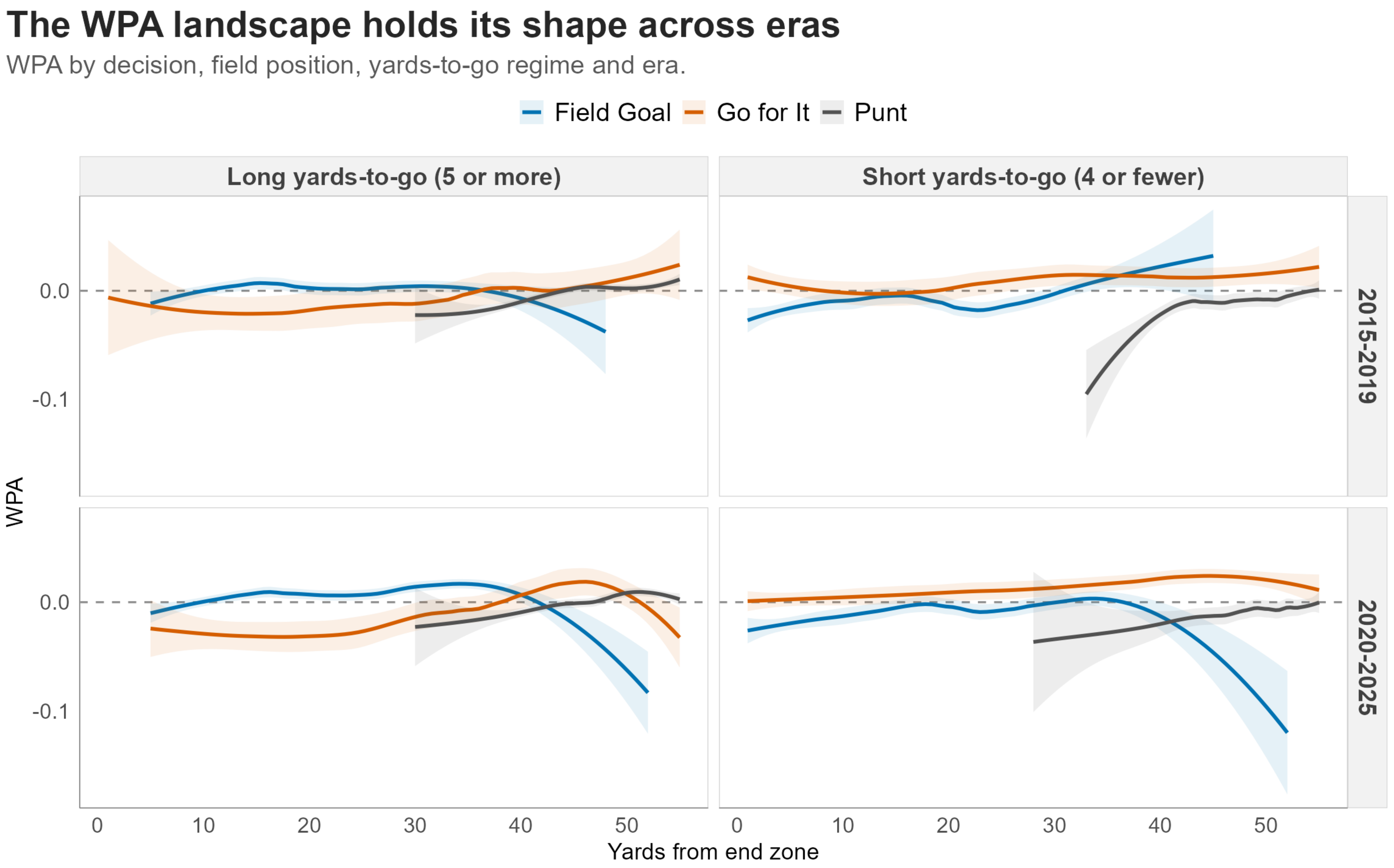
In-sample separation

APPENDIX A11



WPA by regime and era

APPENDIX A12



Open questions

APPENDIX A13

Speculation, not analysis

Why did fourth-down behavior change so sharply around 2018 to 2020? Coaching trees, the spread of analytics staff, and public win-probability tools are all plausible and none of them are tested here.

Whether the shift is better described as a change in beliefs or a change in willingness to act on beliefs already held.

What a causal treatment would need

An estimand defined on the decision itself rather than on the observed attempts.

An identification argument for unmeasured coach and roster factors, which the propensity model conditions on only through their observable proxies.

A defensible answer to what “would have happened” means for a kick that was never taken.